

# Security Engineering with Generative AI

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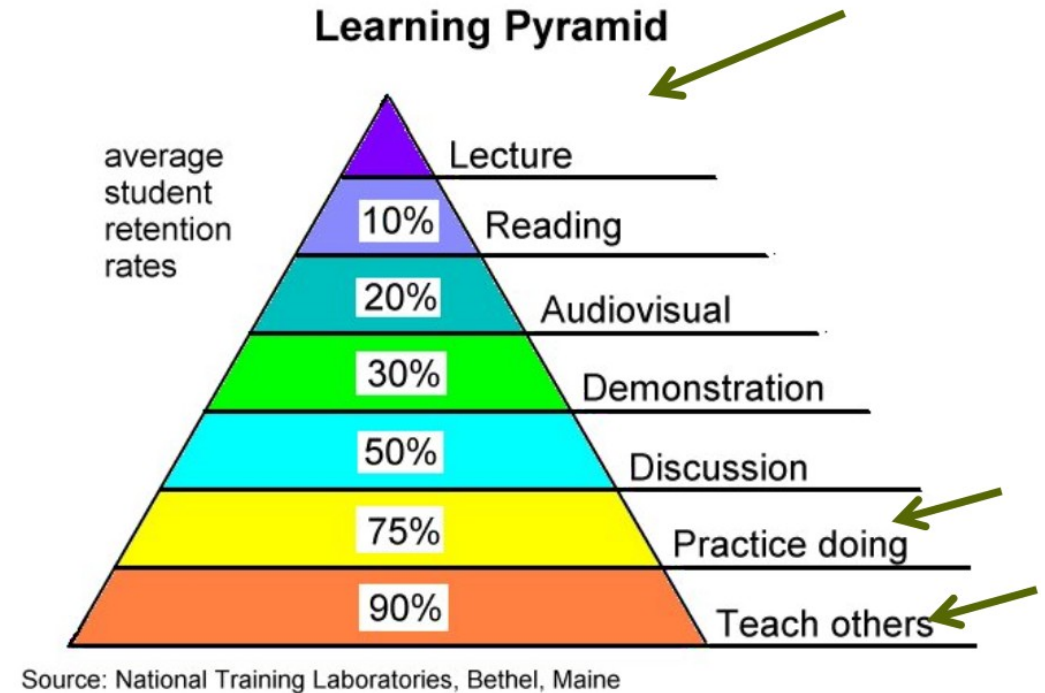


# About the course

- A primer on using generative AI and LLMs
  - A primer on building applications that utilize them
  - A primer on securing generative applications
  - A tour of security problems that can be aided by generative AI
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- Course adopted from [CS 465](#) Generative Security Application Engineering from Portland State University, created by Prof. Wu-chang Feng
    - with NSF funding
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- Moving target: technology evolves very rapidly

# Course Format

- Lecture
  - Hack-a-thon approach each week
  - Topic introduced
  - Lab exercises (started in class)
- Labs & Homework
  - Labs screenshots delivered for each module ☐
  - Individual assignments with screencast walkthroughs ☐
  - Due Mondays



# Assumptions

- Knowledge of Python or willingness to do work outside of class to learn it
- Labs and homeworks all in Python
- Some knowledge of security helpful
  - But will provide content in-place when needed

# Grading

- Attendance and participation
- Lab notebooks PDF for each module, Canvas upload
- Homework programs on Github repository and screencasts (video link)
- Final project
- Cheating policy
  - Homework programs must be developed individually
  - But, use of LLMs mandatory 😊
  - All code submitted must be explained and source cited (e.g. prompt and prompt output must be given)

# Course support

- Canvas
  - Instructor office hours
  - Course resources, schedule, due dates, assignment guidelines, grading
- Homework Q&A Discussion Forum:
  - ask for help and contribute answers

# Ethics

- You will learn techniques and tools for compromising users and systems.
- Do **\*NOT\*** use them against any site outside of those you own or are given permission to access.

# How to Get Started

- Complete these Module 1 labs starting from the *Labs and Homeworks* [main page](#):
  - 01.1 Setup
  - 01.2 Ollama
  - 01.3 Programmatic Model Access
- Install Visual Studio Code (VSCode): follow the steps from the corresponding on Canvas :
  - recommended (**not required**): install an AI Coding Agent (such as OpenAI Codex, Github Copilot, Google Antigravity, Anthropic Claude Code